



Descriptive Statistics Exercises - Solutions

Exercise 2.1.

(a) Find the mean of these numbers:

77 431 875 716 843 601 482 325

The mean is given by the sum of the values divided by the number of values, which is

$$\frac{77 + 431 + 875 + 716 + 843 + 601 + 482 + 325}{8} = \frac{4350}{8} = 543.75$$

(a) Find the mean of this frequency data:

x	0	1	2	3	4	5
Frequency	15	34	94	45	17	35

To find the mean, we must multiply each value by the frequency with which it occurs, then divide by the total number of measurements (which we can find by taking the sum of the frequencies). This is

$$\frac{(0 \times 15) + (1 \times 34) + (2 \times 94) + (3 \times 45) + (4 \times 17) + (5 \times 35)}{15 + 34 + 94 + 45 + 17 + 35} = \frac{600}{240} = 2.5$$

(a) Estimate the mean of this grouped data:

Class	Frequency
$0 \leq x < 10$	42
$10 \leq x < 20$	47
$20 \leq x < 30$	94
$30 \leq x < 50$	35
$50 \leq x \leq 100$	64

To find an estimate for the mean, we need to multiply each frequency value by the midpoint of each class (in this case: 5, 15, 25, 40 and 75). This gives:

$$(5 \times 42) + (15 \times 47) + (25 \times 94) + (40 \times 35) + (75 \times 64) = 9465$$

We can then divide this by the total number of observations, which is the sum of the frequency column. This gives the estimate for the mean as:

$$\frac{9465}{42 + 47 + 94 + 35 + 64} = \frac{9465}{282} = 33.56$$

Exercise 2.2.

- (a) For each of the datasets in Exercise 2.1, find the mode (or modal category), and find or estimate the median and upper/lower quartiles.

(a) The numbers given in increasing order are:

77 325 431 482 601 716 843 875

There are 8 values, so the median will be given by the mean of the fourth and fifth values:

$$Q_2 = \frac{482 + 601}{2} = 541.5$$

The lower and upper quartiles will be given by the means of the 2nd/3rd and 6th/7th values:

$$Q_1 = \frac{325 + 431}{2} = 378 \quad Q_3 = \frac{716 + 843}{2} = 779.5$$

- (b) There are $15 + 34 + 94 + 45 + 17 + 35 = 240$ measurements, so the median will be the mean of the 120th/121st value, and the lower and upper quartiles will be the mean of the 60th/61st and 180th/181st values. By counting from the start of the dataset, these pairs of values fall in the classes with values 2, 2 and 3 respectively, so

$$Q_2 = 2 \quad Q_1 = 2 \quad Q_3 = 3$$

- (c) The total number of measurements is $42 + 47 + 94 + 35 + 64 = 282$, so the median will be given by the mean of the 141st/142nd values, and the quartiles by the 71st and 212th values. By counting through the categories, the median will be in the category $20 \leq x < 30$, so this is the median category. Similarly, Q_1 lies in the category $10 \leq x < 20$, and Q_3 lies in the category $30 \leq x < 50$. If we wanted numerical estimates for the median and quartiles, we could use the midpoints of each category (here, 25, 15 and 40). For more accurate estimates, we could note that the median is the 52nd data point out of 94 in the $20 \leq x < 30$, so instead of taking the midpoint we could estimate it to be $\frac{52}{94}$ of the way through the category:

$$Q_2 \approx 20 + \frac{52}{94} \times (30 - 20) = 25.53$$

Similarly, the quartiles could be estimated as:

$$Q_1 \approx 10 + \frac{29}{47} \times (20 - 10) = 16.17 \quad Q_3 \approx 30 + \frac{29}{35} \times (50 - 30) = 46.5$$

- (b) For the children's heights data from Example 2.3, find the modal category and an estimate of the upper and lower quartiles.

Height (cm)	Midpoint	Frequency (f)
60-80	70	10
80-100	90	12
100-110	105	21
110-120	115	13
120-160	140	4

The category with the largest frequency is 100-110cm. There are a total of $10 + 12 + 21 + 13 + 4 = 60$ data points, so the quartiles lie in the categories containing the 15/16th and 45/46th data points - in this case, 80-100cm and 110-120cm. We could estimate $Q_1 = 90\text{cm}$ and $Q_3 = 115\text{cm}$.

Exercise 2.3.

(a) For each of the datasets in Exercise 2.1, find or estimate the range and interquartile range.

(a) The largest and smallest values are 875 and 77, so the range is $875 - 77 = 798$. The interquartile range is $Q_3 - Q_1 = 779.5 - 378 = 401.5$

(b) The largest and smallest values are 5 and 0, so the range is $5 - 0 = 5$. The interquartile range is $Q_3 - Q_1 = 3 - 2 = 1$.

(c) The largest and smallest possible values are 100 and 0, so the range is $100 - 0 = 100$. An estimate for the interquartile range is $Q_3 - Q_1 = 40 - 15 = 30$, or for our more accurate estimates, $Q_3 - Q_1 = 46.5 - 16.17 = 30.33$.

(b) For the children's heights data from Example ??, using the quartiles you calculated earlier, find the interquartile range.

Using the midpoints of the categories, $115 - 90 = 25\text{cm}$.

Exercise 2.4. For each of the datasets in Exercise 2.1, find or estimate the variance.

(a) The variance is $s_x^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$. For this dataset, we have:

$$\begin{aligned} s_x^2 &= \frac{1}{7} \times ((77 - 543.75)^2 + (431 - 543.75)^2 + (875 - 543.75)^2 + (716 - 543.75)^2 \\ &\quad + (843 - 543.75)^2 + (601 - 543.75)^2 + (482 - 543.75)^2 + (325 - 543.75)^2) \\ &= \frac{1}{7} \times ((-466.75)^2 + (218.75)^2 + (-112.75)^2 + (-61.75)^2 + (57.25)^2 + (172.25)^2 + (299.25)^2 + (331.25)^2) \\ &= \frac{1}{7} \times 514457.5 = 73493.9 \end{aligned}$$

(b) Using the other formula for the variance:

$$s_x^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

There are 240 measurements, so $n - 1 = 239$. The sum of the values we calculated earlier to be 600. The sum of the squares is given by:

$$\sum x^2 = (15 \times 0^2) + (34 \times 1^2) + (94 \times 2^2) + (45 \times 3^2) + (17 \times 4^2) + (35 \times 5^2) = 1962$$

So using the formula:

$$s_x^2 = \frac{1}{239} \left(1962 - \frac{600^2}{240} \right) = 1.933$$

(c) There are 282 measurements, so $n - 1 = 281$. The sum of the values (assuming each is at the midpoint of the category) we calculated earlier to be 9465. The sum of the squares is given by:

$$\sum x^2 = (5^2 \times 42) + (15^2 \times 47) + (25^2 \times 94) + (40^2 \times 35) + (75^2 \times 64) = 486375$$

So using the formula:

$$s_x^2 = \frac{1}{281} \left(486375 - \frac{9465^2}{282} \right) = 600.33$$

Exercise 2.5.

(a) For each of the datasets in Exercise 2.1, find or estimate the standard deviation.

(a) The variance was calculated to be 73493.9, so the standard deviation is:

$$s_x = \sqrt{73493.9} = 271.1$$

(b) The variance was calculated to be 1.933, so the standard deviation is:

$$s_x = \sqrt{1.933} = 1.39$$

(c) The variance was calculated to be 600.33, so the standard deviation is:

$$s_x = \sqrt{600.33} = 24.5$$

(b) For the children's heights data from Example 2.3, find an estimate of the standard deviation.

There are 60 measurements, so $n - 1 = 59$. The sum of the values (assuming each is at the midpoint of the category) we calculated earlier to be 6040. The sum of the squares is given by:

$$\sum x^2 = (70^2 \times 10) + (90^2 \times 12) + (105^2 \times 21) + (115^2 \times 13) + (140^2 \times 4) = 628050$$

So using the formula:

$$s_x^2 = \frac{1}{59} \left(628050 - \frac{6040^2}{60} \right) = 339.4$$

Then $s_x = \sqrt{339.4} = 18.42$

Exercise 2.6.

Which of the data points on the plot above are outliers, by the IQR definition, and by the standard deviation definition?

The estimated IQR for the categorical data is 25 (the actual IQR for the points plotted is 20.6), and the mean is 100.7, so 1.5 IQRs either side of the mean are the points $100.7 + 25 = 125.7$ and $100.7 - 25 = 75.7$. Based on this, we could consider any values less than 75.7 or more than 125.7 to be outliers.

The standard deviation was estimated to be 18.42 (the actual SD for the points plotted is 18.75), so by the 'three standard deviations' definition we could consider values more than $100.7 + (3 \times 18.42) = 155.96$ and less than $100.7 - (3 \times 18.42) = 45.44$ to be outliers.